Chapter 8

Bipolar Junction Transistors (BJT)

8.1 Introduction

The invention of the transistor in 1947 by Shockley, Bardeen, and Brattain at Bell Labs revolutionized the world. Before the transistor, electronics relied on bulky, fragile, and power-hungry vacuum tubes. The transistor allowed for the miniaturization of electronics, leading to the computers, smartphones, and automated agricultural systems we rely on today.

The **Bipolar Junction Transistor (BJT)** is a three-terminal semiconductor device. The term ”Bipolar” indicates that both electrons and holes are involved in the conduction process. It is primarily used for two functions:

1. **Amplification:** Increasing the amplitude of a weak signal (e.g., amplifying sensor data from a soil moisture probe).
2. **Switching:** Turning a current ON or OFF electronically (e.g., digital logic, controlling a relay or solenoid valve).

Learning Objectives

By the end of this chapter, you should be able to:

- Describe the physical construction of NPN and PNP transistors.
- Explain the "Transistor Action" using the water valve analogy.
- Define the role of the Emitter, Base, and Collector.
- Derive the fundamental relationship: $I_E = I_B + I_C$.
- Calculate DC Current Gains α and β.
- Analyze the impact of Leakage Current (I_{CBO} and I_{CEO}).

8.2 Transistor Construction

A BJT consists of three doped semiconductor regions separated by two PN junctions. There are two types of BJTs:

8.2.1 NPN Transistor

A thin layer of P-type material (Base) is sandwiched between two N-type materials (Emitter and Collector). This is the most common type because electrons (the majority carriers) have higher mobility than holes, making NPN transistors faster.

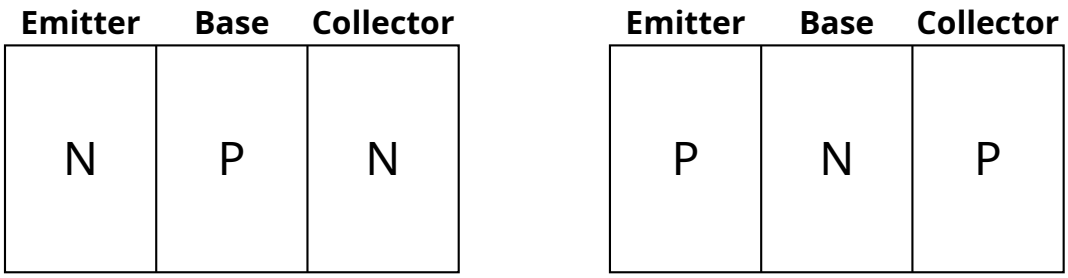
8.2.2 PNP Transistor

A thin layer of N-type material (Base) is sandwiched between two P-type materials.

8.3 The Terminals

The three regions have distinct doping levels and physical sizes:

1. **Emitter (E):**
   - **Doping:** Heavily doped.



(a) NPN Structure **(b) PNP Structure**

Figure 8.1: Block diagrams of NPN and PNP Transistors.

- **Function:** To inject (emit) a large number of majority carriers (electrons in NPN) into the Base.

2. **Base (B):**
   - **Doping:** Lightly doped.
   - **Physical Size:** Very thin ($< 1\mu m$).
   - **Function:** To pass most of the carriers from Emitter to Collector while controlling the flow via recombination.
3. **Collector (C):**
   - **Doping:** Moderately doped.
   - **Physical Size:** The largest region (to dissipate heat).
   - **Function:** To collect the carriers coming from the Base.

8.4 Transistor Operation (Active Region)

To use the transistor as an amplifier, we must bias it in the **Active Region**.

- The **Base-Emitter (BE)** junction must be **Forward Biased**.
- The **Collector-Base (CB)** junction must be **Reverse Biased**.

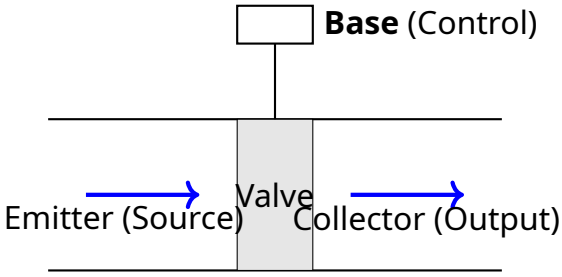
8.4.1 Operation of NPN Transistor

1. **Injection:** The forward bias at the BE junction pushes electrons from the N-type Emitter into the P-type Base.

2. **Diffusion & Recombination:** The Base is very thin and lightly doped. Only a tiny fraction (less than 5%) of the electrons recombine with holes in the Base to form the Base Current (I_B).

3. **Collection:** The remaining electrons (more than 95%) diffuse across the Base and reach the depletion region of the CB junction. Since the CB junction is reverse biased with a high positive voltage at the Collector, these electrons are swept rapidly across into the Collector, forming the Collector Current (I_C).

This mechanism is called **Transistor Action**: A small current in the Base controls a large current flowing from Emitter to Collector.



Water Analogy: Small turn of handle controls huge flow.

Figure 8.2: The Transistor as a Valve.

8.5 Circuit Symbols and Currents

The arrow on the schematic symbol defines the emitter and the direction of conventional current flow.

- **NPN:** Arrow points **OUT** (Not Pointing iN).
- **PNP:** Arrow points **IN** (Pointing iN).

8.5.1 The Fundamental Current Equation

Applying Kirchhoff's Current Law (KCL) to the transistor as a single node:

$$I_E = I_C + I_B \tag{8.1}$$

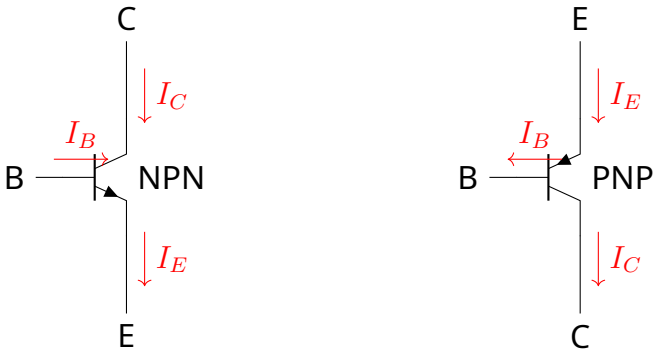


Figure 8.3: Circuit Symbols and Conventional Current Directions.

Since the Base current is very small ($I_B \ll I_C$), we often approximate:

$$I_E \approx I_C$$

8.6 Transistor Current Gains

We define two major parameters to quantify the amplification capability of a transistor.

8.6.1 DC Alpha (α_{dc})

Alpha relates the Collector current to the Emitter current. It represents the efficiency of carrier transfer from Emitter to Collector.

$$\alpha_{dc} = \frac{I_C}{I_E} \tag{8.2}$$

Since $I_C < I_E$ (due to recombination loss in the Base), α is always less than 1.

- Typical Range: 0.95 to 0.998.
- Ideally: $\alpha = 1$.

8.6.2 DC Beta (β_{dc})

Beta relates the Collector current to the input Base current. This is the **Amplification Factor**.

$$\beta_{dc} = \frac{I_C}{I_B} \tag{8.3}$$

This parameter is often labeled as h_{FE} on datasheets.

- Typical Range: 50 to 400.
- Meaning: A β of 100 means that for every 1 electron recombining in the base, 100 electrons reach the collector.

8.6.3 Relationship between α and β

We can derive the relationship using $I_E = I_C + I_B$. Divide by I_C:

$$\frac{I_E}{I_C} = 1 + \frac{I_B}{I_C}$$

$$\frac{1}{\alpha} = 1 + \frac{1}{\beta} = \frac{\beta + 1}{\beta}$$

Therefore:

$$\alpha = \frac{\beta}{\beta + 1} \tag{8.4}$$

$$\beta = \frac{\alpha}{1 - \alpha} \tag{8.5}$$

Example 1: Gain Calculation

A transistor has a Collector current of 10 mA and a Base current of 40 μA. Calculate β and α.

Solution: Convert units: $I_B = 40\mu\text{A} = 0.04$ mA.

$$\beta = \frac{I_C}{I_B} = \frac{10 \text{ mA}}{0.04 \text{ mA}} = 250$$

$$\alpha = \frac{\beta}{\beta + 1} = \frac{250}{251} = 0.996$$

8.7 Leakage Current

In the ideal explanation, I_C consists only of carriers injected from the Emitter. However, the Collector-Base junction is a reverse-biased diode. Just like any diode, it has a small **Reverse Saturation Current** due to minority carriers.

8.7.1 I_{CBO} (Collector-Base Open)

This is the leakage current flowing from Collector to Base with the Emitter **Open** ($I_E = 0$). It is very small (nA) and temperature dependent.

8.7.2 I_{CEO} (Collector-Emitter Open)

This is the leakage flowing from Collector to Emitter with the Base **Open** ($I_B = 0$). Due to the transistor action, this leakage is amplified by the transistor itself!

$$I_{CEO} = (\beta + 1)I_{CBO} \approx \beta I_{CBO} \tag{8.6}$$

While I_{CBO} is tiny, I_{CEO} can be significant, especially in Germanium transistors or at high temperatures.

General Collector Current Equation: Taking leakage into account, the total collector current is:

$$I_C = \beta I_B + I_{CEO} \tag{8.7}$$

For modern silicon transistors at room temperature, I_{CEO} is negligible, so $I_C \approx \beta I_B$ holds true.

8.8 Chapter Summary

1. The BJT is a current-controlled device consisting of Emitter, Base, and Collector.
2. **NPN** is preferred over PNP due to higher electron mobility.
3. For amplification, the B-E junction is forward biased, and the C-B junction is reverse biased.
4. $I_E = I_C + I_B$.
5. $\beta = I_C/I_B$ is the current gain (typically 100).
6. $\alpha = I_C/I_E$ is the transfer efficiency (typically 0.99).

8.9 Review Questions

Part A: Conceptual Questions

1. Explain why the Base region of a BJT must be very thin and lightly doped.
2. Define ”Transistor Action” and explain how a small base current controls a large collector current.
3. Why is the Collector region physically larger than the Emitter and Base regions?
4. Describe the biasing conditions required for a transistor to operate in the (a) Active Region, (b) Saturation Region, and (c) Cutoff Region.
5. Differentiate between I_{CBO} and I_{CEO}. Which one is larger and why?
6. Draw the circuit symbols for NPN and PNP transistors and indicate the direction of all currents (I_E, I_B, I_C).
7. Why are NPN transistors more commonly used than PNP transistors?
8. What happens to the current gain β if the temperature of the transistor increases?
9. Define ”Punch-through” or ”Reach-through” in a BJT.
10. Can a transistor be used as a diode? If so, how would you connect the terminals?

Part B: Advanced Calculations & Analysis

11. **Alpha-Beta Conversion:** A transistor has $\alpha = 0.98$.
    (a) Calculate β.
    (b) If $I_E = 5$ mA, calculate I_C and I_B.
12. **Beta Range:** A transistor datasheet specifies β ranges from 100 to 300. Calculate the range of α for this transistor. Explain what this implies about the efficiency of the transistor.

13. **Current Calculation:** Measurements on an NPN transistor circuit show $I_B = 20\mu$A and $I_E = 2.5$ mA.
    (a) Calculate I_C.
    (b) Calculate β and α.

14. **Leakage Calculation:** A Germanium transistor has a leakage current $I_{CBO} = 5\mu$A and $\alpha = 0.99$.
    (a) Calculate I_{CEO}.
    (b) If the base is left open ($I_B = 0$), what current flows through the collector?

15. **Sensitivity Analysis:** A transistor has a $\beta = 100$. Due to temperature rise, α increases by only 0.5% (from its initial value).
    (a) Calculate the initial α.
    (b) Calculate the new α and the corresponding new β.
    (c) What is the percentage change in β? (Comment on the stability of β vs α).